EDA — Variation

PSY 410: Data Science for Psychology

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What is EDA?

Know your data before you test it

Before you run a single statistical test, you should know your data well enough to predict what the results will look like.

That's what EDA gives you — **no surprises**.

It's also how you catch problems early: rows that didn't load, codes that weren't recoded, "999" sneaking in as a real age.

Exploratory vs. confirmatory

	Exploratory (EDA)	Confirmatory
Goal	Discover patterns	Test hypotheses
Attitude	Curiosity	Rigor
Questions	Open-ended	Pre-specified
Output	Visualizations, hunches	p-values, conclusions

The two roles get mixed up if you do them on the same data. **Preregistration** keeps them honest: you write your confirmatory hypotheses down *before* you peek.

A cleaner pattern: split your data

If you're worried about exploring and testing on the same dataset, use multiple datasets:

- **Dataset 1** → EDA. Look around. Generate ideas.
- **Dataset 2** → Confirm ideas from Set 1. Then EDA on Set 2 → new ideas.
- **Dataset 3** → Confirm ideas from Sets 1 & 2. Then EDA on Set 3 → new ideas.
- … and so on.

Each new dataset gets one shot at confirmation, then becomes fair game for exploration.

Good EDA means looking before you test

> "EDA is an attitude, not a technique." — John Tukey

- Ask questions
- Answer them by visualizing data
- Use what you learn to ask new questions
- Repeat

There is no single "right" way to do EDA. The goal is **understanding.**

What we're doing today

Exploring **variation** — what does a single variable look like?

- Distribution shape
- Center and spread
- Outliers and unusual values
- Missing data

Tomorrow: **covariation** — how do variables relate to each other?

Our dataset: Big Five Personality

```r
dim(bfi)
```

```
[1] 2800   28
```

```r
names(bfi)
```

```
 [1] "A1"        "A2"        "A3"        "A4"        "A5"        "C1"
 [7] "C2"        "C3"        "C4"        "C5"        "E1"        "E2"
[13] "E3"        "E4"        "E5"        "N1"        "N2"        "N3"
[19] "N4"        "N5"        "O1"        "O2"        "O3"        "O4"
[25] "O5"        "gender"    "education" "age"
```

25 items measuring five personality factors (Agreeableness, Conscientiousness, Extraversion, Neuroticism, Openness) + demographics.

What’s in here

Variable	Description
A1–A5	Agreeableness items (1–6 scale)
C1–C5	Conscientiousness items (1–6 scale)
E1–E5	Extraversion items (1–6 scale)
N1–N5	Neuroticism items (1–6 scale)
O1–O5	Openness items (1–6 scale)
gender	1 = male, 2 = female (binary coding)
education	1–5 (HS incomplete through graduate)
age	Age in years

Exploring distributions

Start simple: summary statistics

```
bfi |>
  select(age, gender, education) |>
  summary()
```

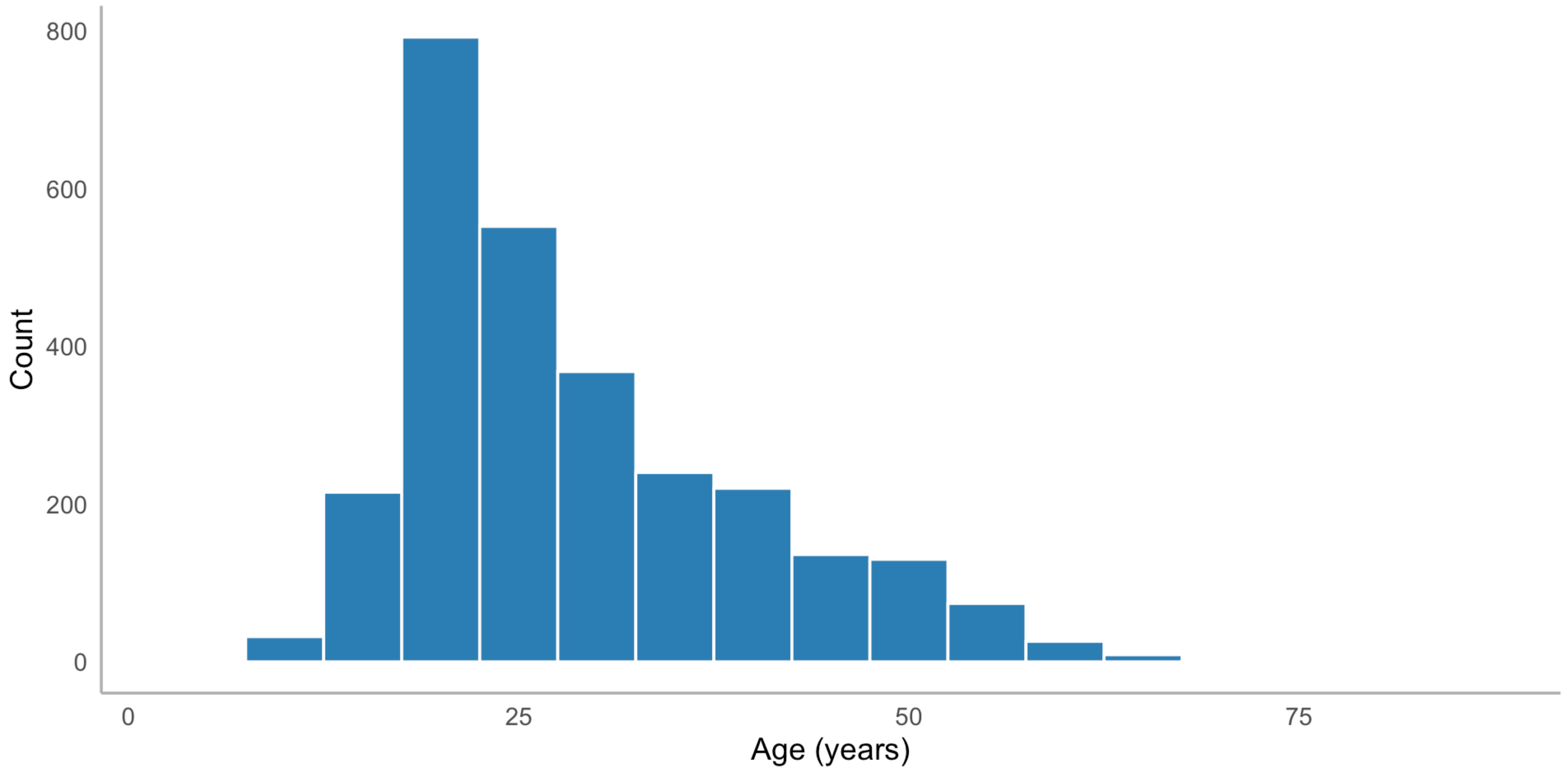
```
      age            gender        education    
 Min.   : 3.00   Min.   :1.000   Min.   :1.00  
 1st Qu.:20.00   1st Qu.:1.000   1st Qu.:3.00  
 Median :26.00   Median :2.000   Median :3.00  
 Mean   :28.78   Mean   :1.672   Mean   :3.19  
 3rd Qu.:35.00   3rd Qu.:2.000   3rd Qu.:4.00  
 Max.   :86.00   Max.   :2.000   Max.   :5.00  
                                 NA's   :223   
```

But summary stats can hide a lot. Always visualize.

Histograms: the workhorse

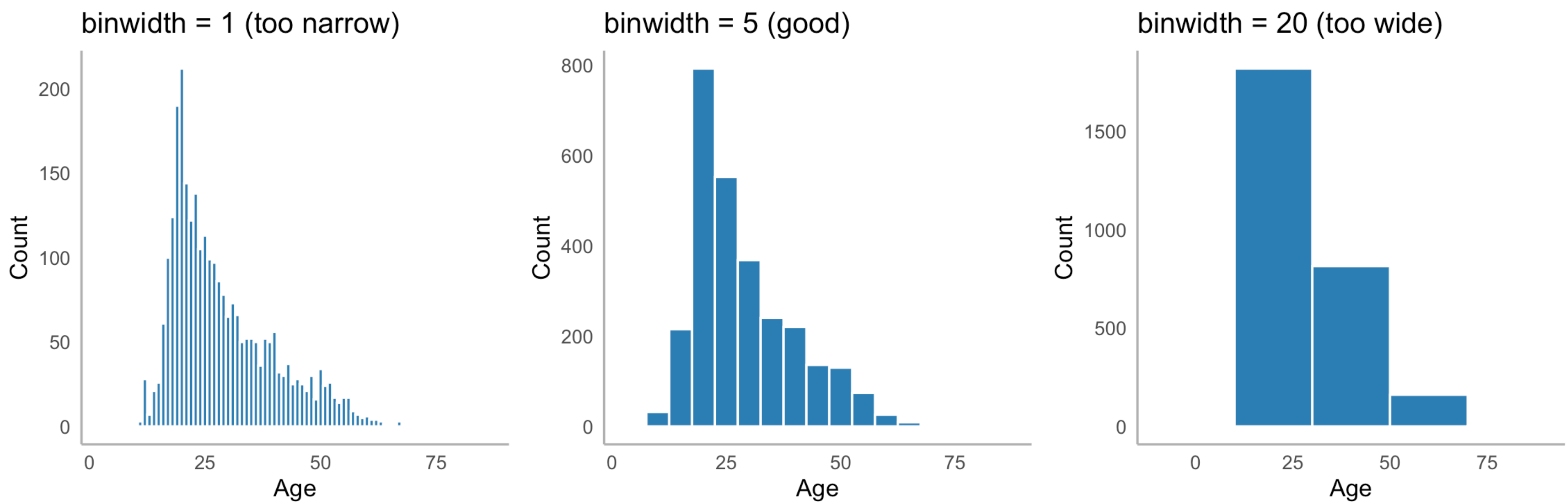
```
bfi |>
  ggplot(aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "#2c7fb8", color = "white") +
  labs(x = "Age (years)", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Histograms: the workhorse



Choosing binwidth

The binwidth matters a lot:

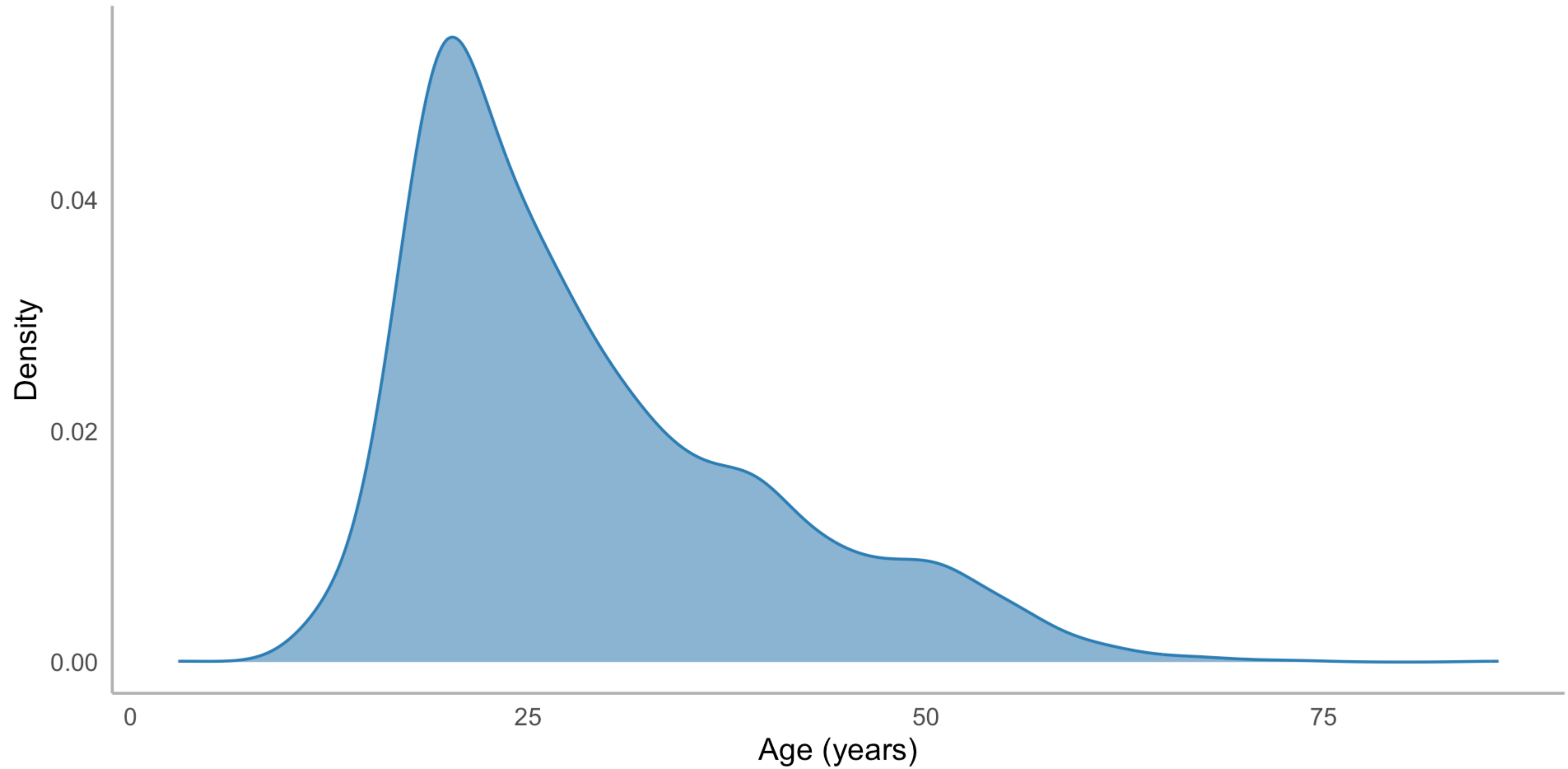


There's no "right" answer — try a few and pick the one that shows the shape clearly.

Density plots: smooth alternative

```r
bfi |>
  ggplot(aes(x = age)) +
  geom_density(fill = "#2c7fb8", alpha = 0.6, color = "#2c7fb8") +
  labs(x = "Age (years)", y = "Density") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Density plots: smooth alternative



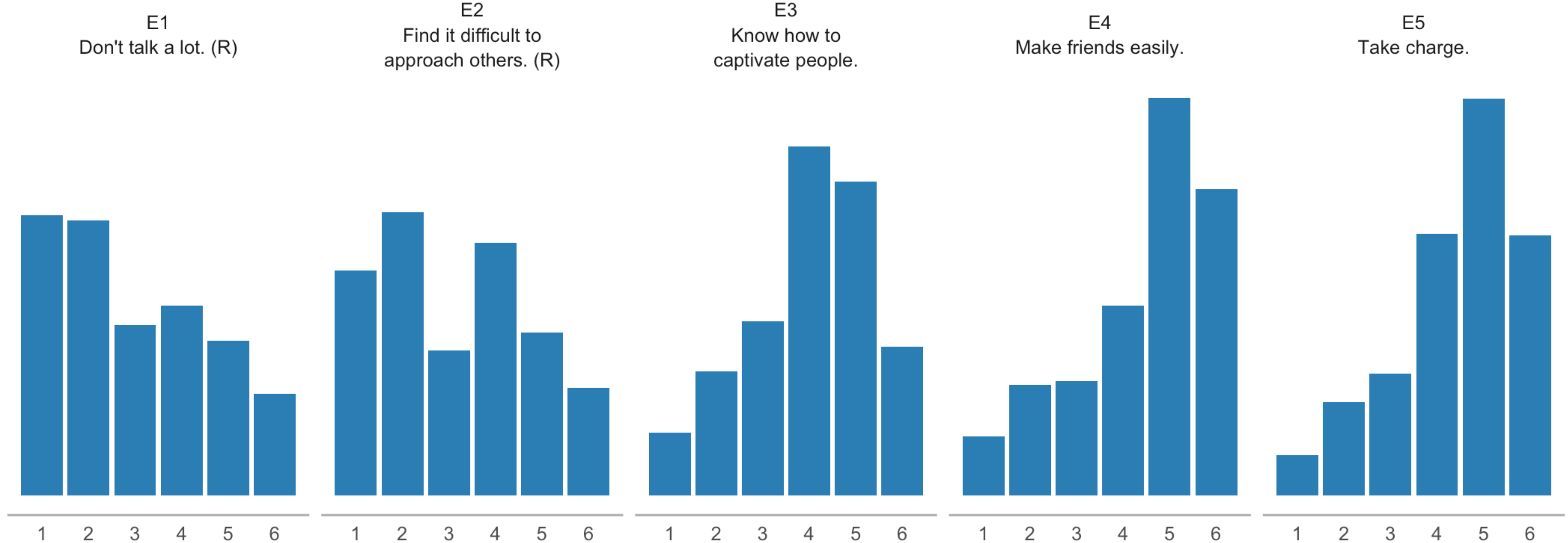
Exploring survey items

```r
# All Extraversion items
item_labels <- c(
  E1 = "E1\nDon't talk a lot. (R)",
  E2 = "E2\nFind it difficult to\napproach others. (R)",
  E3 = "E3\nKnow how to\ncaptivate people.",
  E4 = "E4\nMake friends easily.",
  E5 = "E5\nTake charge."
)

bfi |>
  select(E1:E5) |>
  pivot_longer(everything(), names_to = "item", values_to = "response") |>
  ggplot(aes(x = response)) +
  geom_bar(fill = "#2c7fb8") +
  facet_wrap(~ item, nrow = 1, labeller = as_labeller(item_labels)) +
```

Exploring survey items

Distribution of responses to Extraversion items



Scale: 1 = Very inaccurate, 6 = Very accurate. (R) = reverse-scored.

A quick aside: `factor()`

Education is stored as numbers (`1`–`5`), but the numbers are *codes*, not measurements. `factor()` turns them into ordered categories with readable labels:

```
factor(education,
       levels = 1:5,
       labels = c("Some HS", "HS grad", "Some college",
                  "College grad", "Graduate"))
```

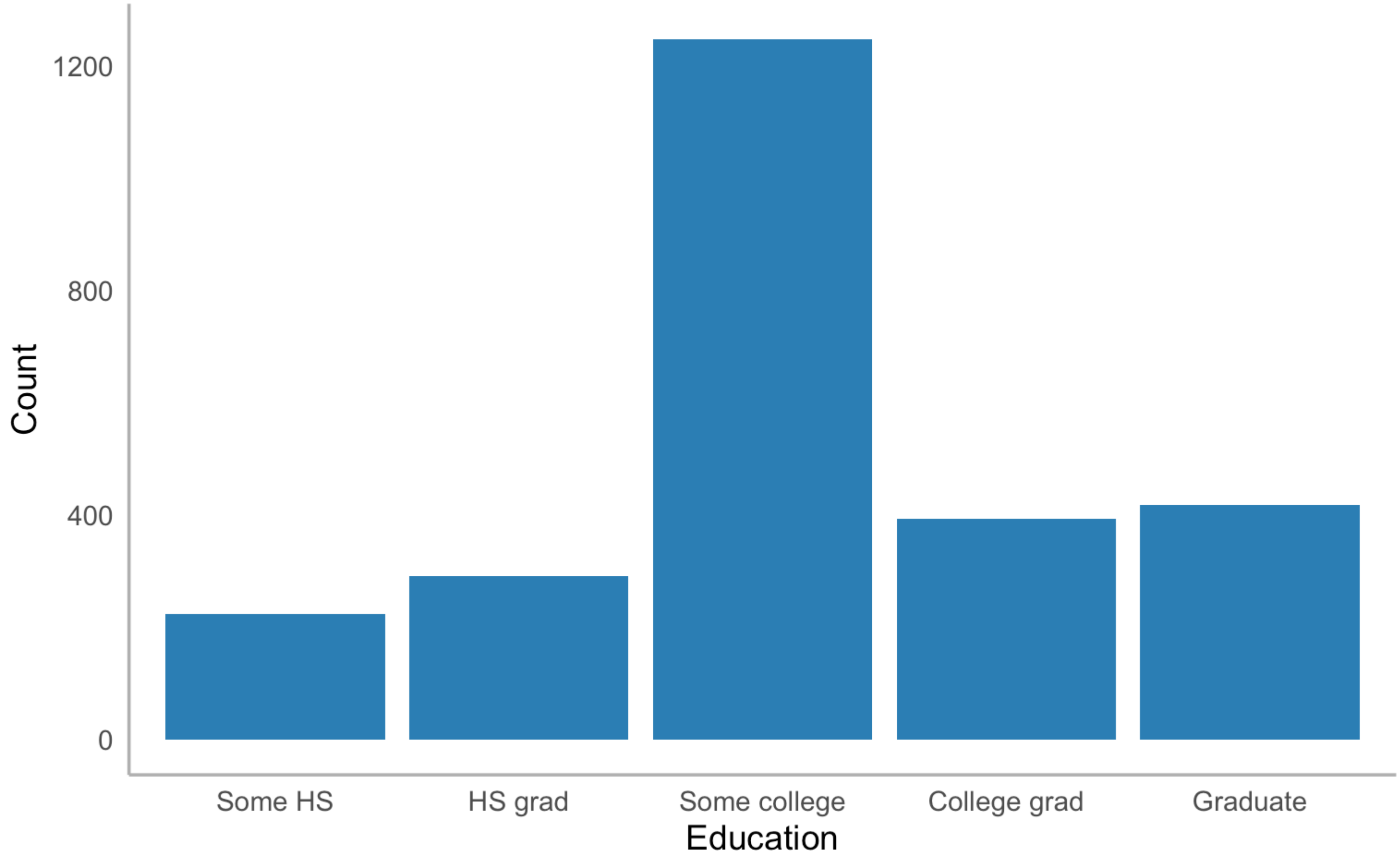
- `levels` = the values currently in the data (in the order you want them)
- `labels` = what to show instead

We'll come back to factors in detail in a later session — for now, just know this is how we attach labels to numeric codes.

Categorical variables

```
bfi |>
  filter(!is.na(education)) |>
  mutate(education = factor(education,
    levels = 1:5,
    labels = c("Some HS", "HS grad", "Some college", "College grad", "Graduat
  ggplot(aes(x = education)) +
  geom_bar(fill = "#2c7fb8") +
  labs(x = "Education", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Categorical variables



Counts and proportions

```
# Counts
bfi |>
  filter(!is.na(education)) |>
  mutate(education = factor(education,
    levels = 1:5,
    labels = c("Some HS", "HS grad", "Some college", "College grad", "Graduat
  count(education)
```

```
# A tibble: 5 × 2
  education        n
  <fct>        <int>
1 Some HS        224
2 HS grad        292
3 Some college  1249
4 College grad   394
5 Graduate       418
```

Counts and proportions

```r
# Proportions
bfi |>
  filter(!is.na(education)) |>
  mutate(education = factor(education,
    levels = 1:5,
    labels = c("Some HS", "HS grad", "Some college", "College grad", "Graduat
  count(education) |>
  mutate(proportion = n / sum(n))
```

```
# A tibble: 5 × 3
  education        n proportion
  <fct>        <int>      <dbl>
1 Some HS        224     0.0869
2 HS grad        292     0.113
3 Some college  1249     0.485
4 College grad   394     0.153
5 Graduate       418     0.162
```

Don’t round inside the data frame

Keep the full precision in your data. Round only when you’re **displaying** it:

```
bfi |>
  count(education) |>
  mutate(proportion = n / sum(n)) |>
  knitr::kable(digits = 2)
```

If you round in the data frame, you’ve thrown away precision you can’t get back. If you round at display, the underlying data is still exact.

Outliers

What are outliers?

Values that are unusual compared to the rest of the data. They might be:

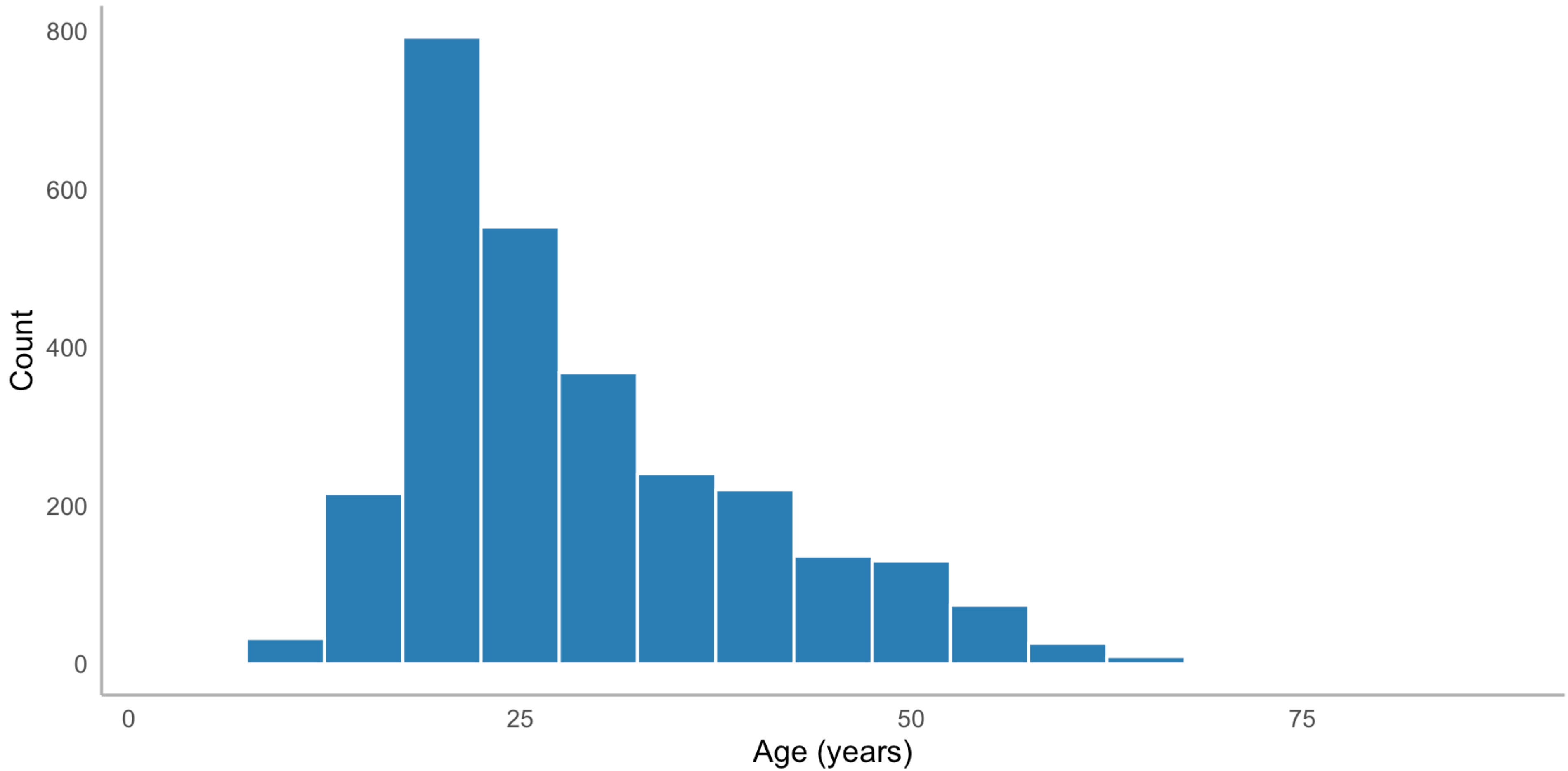
- **Real** — genuinely extreme values (a 95-year-old in a college study)
- **Errors** — data entry mistakes (age = 999)
- **Interesting** — worth investigating

Never delete an outlier without understanding it.

Spotting outliers visually

```r
bfi |>
  ggplot(aes(x = age)) +
  geom_histogram(binwidth = 5, fill = "#2c7fb8", color = "white") +
  labs(x = "Age (years)", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Spotting outliers visually

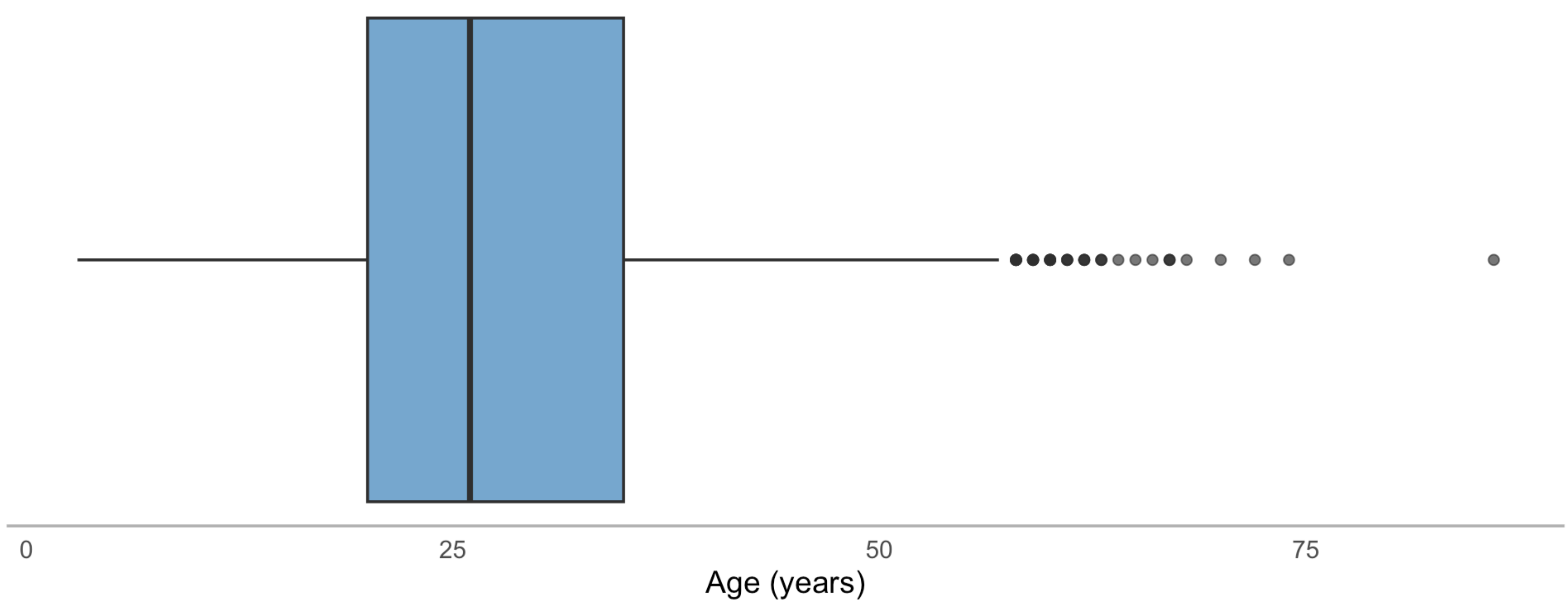


Spotting outliers with boxplots

```r
bfi |>
  ggplot(aes(x = age)) +
  geom_boxplot(fill = "#2c7fb8", alpha = 0.7) +
  labs(x = "Age (years)") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line.x = element_line(color = "gray70"),
        axis.text.y = element_blank(),
        axis.ticks.y = element_blank())
```

Points beyond the whiskers are flagged as outliers by the 1.5×IQR rule.

Spotting outliers with boxplots



Investigating outliers programmatically

```r
# Find unusually old or young participants
bfi |>
  filter(age > 60) |>
  select(age, gender, education)
```

```
# A tibble: 22 × 3
     age gender education
   <int>  <int>     <int>
 1    68      1         5
 2    64      2         5
 3    74      1         5
 4    63      2         3
 5    62      1         2
 6    86      2         2
 7    61      1         2
 8    67      1         5
 9    67      1         5
```

What to do with outliers

Outlier type	What to do
Likely error (age = 999)	Fix or set to NA
Real but extreme	Note it, keep it, mention in write-up
Suspicious	Investigate further
Doesn't affect conclusions	Keep it, move on

Document your decisions. Future you will want to know.

Let’s explore YOUR data

Our class survey

Remember that survey you filled out Week 1? Let's do some EDA on it.

```
class_survey <- read_csv(
  "https://raw.githubusercontent.com/sjweston/datasci410/refs/heads/main/data
)
```

```
glimpse(class_survey)
```

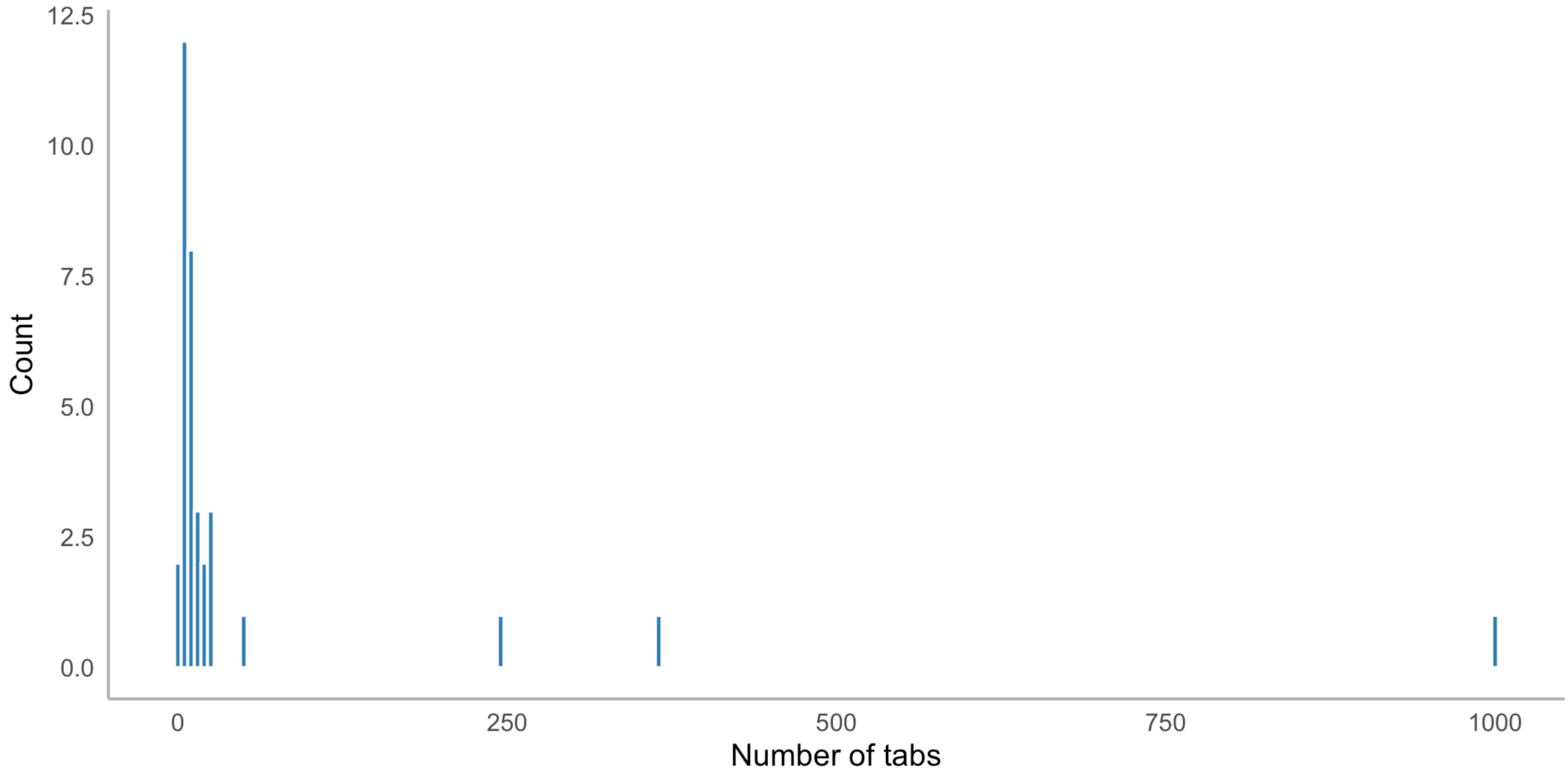
```
Rows: 34
Columns: 15
$ response_id       <chr> "R_7Fkq5bRdEYWENih", "R_7mni7E59hbZtnha",
"R_7BKhLj1…
$ major             <chr> "Psychology", "Neuroscience", "Psychology",
"Psychol…
$ year              <chr> "Junior", "Senior", "Junior", "Junior", "Senior",
"S…
$ chronotype        <chr> "Evening person", "Morning person", "Morning
person"…
```

```
$ sleep_hrs         <dbl> 8.0, 6.0, 9.0, 9.0, 7.0, 7.0, 7.0, 7.0, 7.0, 7.0,
7.…
$ caffeine_per_day  <dbl> 2.00, 1.00, 2.00, 1.00, 2.00, 1.00, 0.15, 1.50,
1.30…
```

How many tabs do you have open?

```r
ggplot(class_survey, aes(x = tabs_open)) +
  geom_histogram(binwidth = 5, fill = "#2c7fb8", color = "white") +
  labs(x = "Number of tabs", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

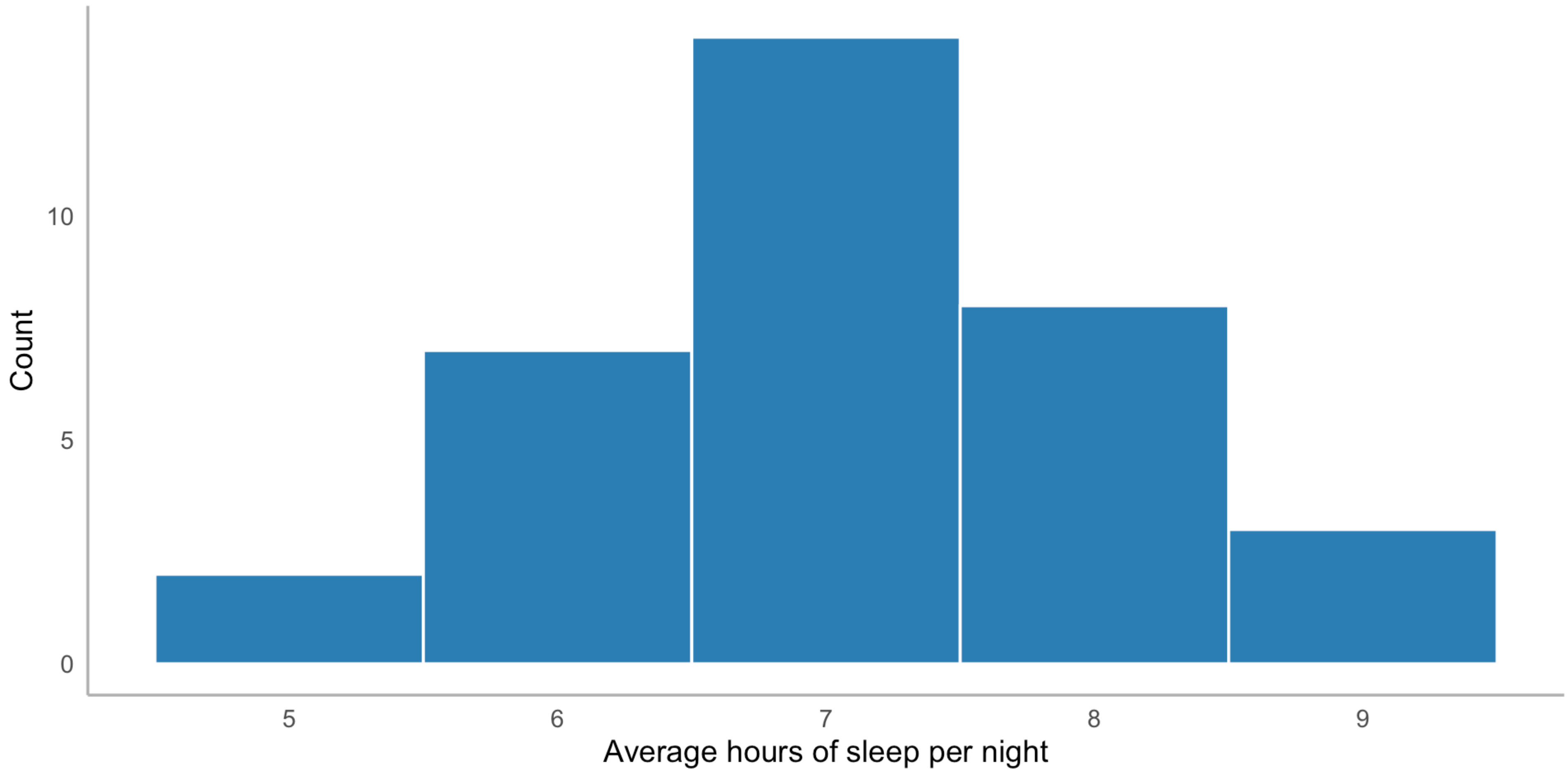
How many tabs do you have open?



How does our class sleep?

```
ggplot(class_survey, aes(x = sleep_hrs)) +
  geom_histogram(binwidth = 1, fill = "#2c7fb8", color = "white") +
  labs(x = "Average hours of sleep per night", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

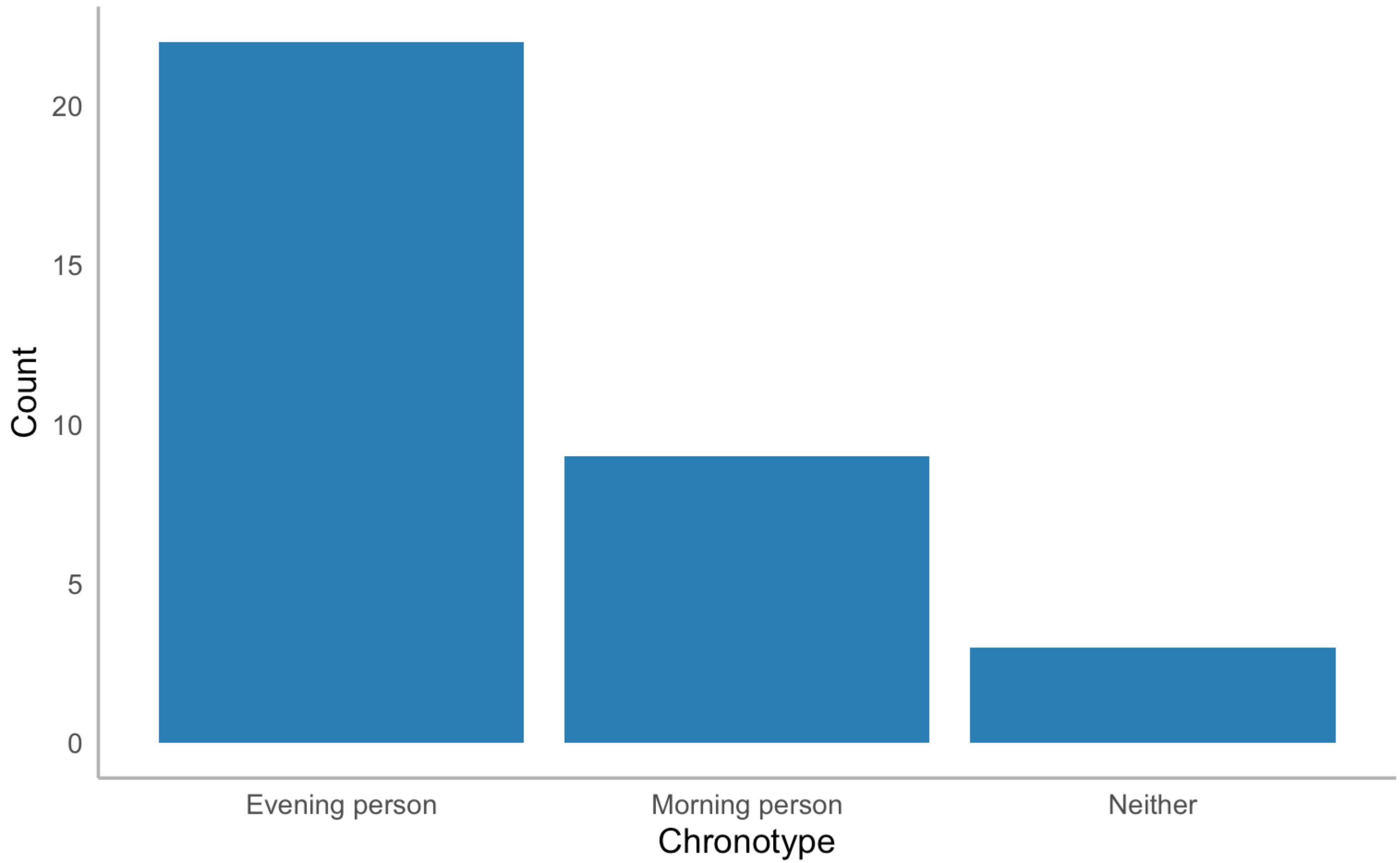
How does our class sleep?



Morning people or night owls?

```
ggplot(class_survey, aes(x = chronotype)) +
  geom_bar(fill = "#2c7fb8") +
  labs(x = "Chronotype", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Morning people or night owls?



Pair coding break

Your turn: 10 minutes

Using the `class_survey` data, look at the distribution of **`social_media_hrs`** (self-reported hours per day on social media):

1. Plot the distribution — what shape is it?
2. Check for **outliers**. Are any values suspicious?
3. What does this tell you about our class?

> **Tip**
>
> Try both a histogram and a boxplot. They show different things. Use `summary(class_survey$social_media_hrs)` first to get oriented.

Before we move on

📤 Submit your code on Canvas for participation credit. Paste what you have — it doesn't need to work perfectly.

Missing data — a first look

Missing data are everywhere

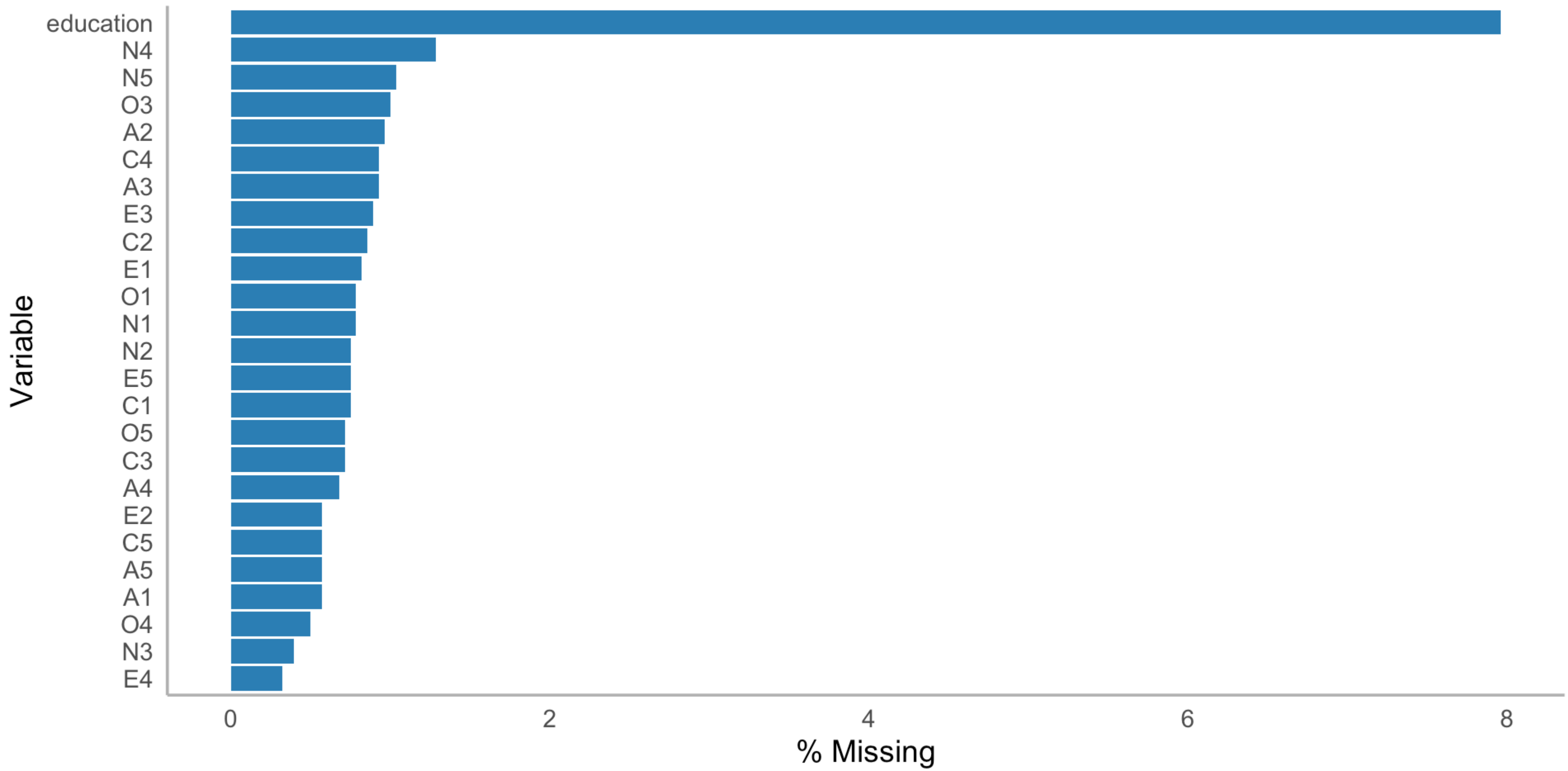
```r
# How much missing data do we have?
bfi |>
  summarize(across(everything(), ~ sum(is.na(.)))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing")
  arrange(desc(n_missing))
```

```
# A tibble: 28 × 2
   variable  n_missing
   <chr>         <int>
 1 education       223
 2 N4               36
 3 N5               29
 4 O3               28
 5 A2               27
 6 A3               26
 7 C4               26
 8 E3               25
 9 C2               24
```

Visualizing missingness

```
bfi |>
  summarize(across(everything(), ~ mean(is.na(.)))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "pct_missing"
  mutate(pct_missing = pct_missing * 100) |>
  filter(pct_missing > 0) |>
  arrange(desc(pct_missing)) |>
  ggplot(aes(x = reorder(variable, pct_missing), y = pct_missing)) +
  geom_col(fill = "#2c7fb8") +
  coord_flip() +
  labs(x = "Variable", y = "% Missing") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"))
```

Visualizing missingness



Why it matters for EDA

Missing data can bias your exploration:

- If missingness is related to the variables you're studying, your distributions are wrong
- If certain groups are more likely to have missing data, you might miss important patterns

We'll cover this in depth in Session 14. For now: **always check for missing data before exploring.**

The EDA workflow

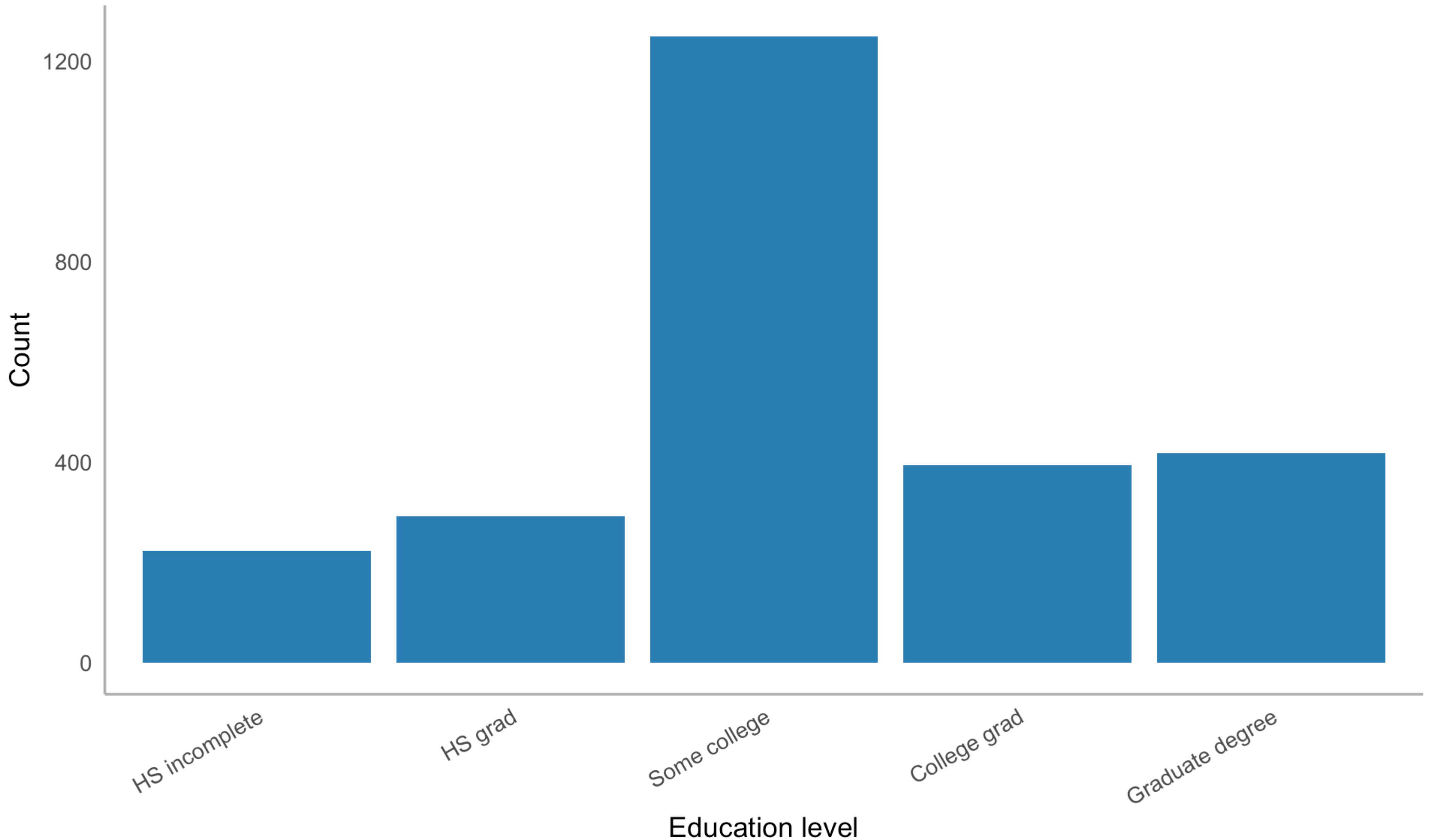
A systematic approach

1. **Look at the structure** — `glimpse()`, `summary()`
2. **Check for missing data** — how much? Which variables?
3. **Explore each variable** — histograms, bar charts, summaries
4. **Look for outliers** — boxplots, programmatic checks
5. **Ask questions** — what surprised you? What do you want to know more about?
6. **Repeat** — EDA is iterative

Putting it together: a mini-EDA

```r
# Education distribution — what does it look like?
bfi |>
  filter(!is.na(education)) |>
  mutate(education = factor(education, labels = c(
    "HS incomplete", "HS grad", "Some college",
    "College grad", "Graduate degree"
  ))) |>
  ggplot(aes(x = education)) +
  geom_bar(fill = "#2c7fb8") +
  labs(x = "Education level", y = "Count") +
  theme_minimal(base_size = 14) +
  theme(panel.grid = element_blank(),
        axis.line = element_line(color = "gray70"),
        axis.text.x = element_text(angle = 30, hjust = 1))
```

Putting it together: a mini-EDA



Wrapping up

EDA toolkit so far

Tool	When to use
`glimpse()`	First look at structure
`summary()`	Quick numeric summaries
`geom_histogram()`	Continuous distributions
`geom_density()`	Smooth distribution shape
`geom_bar()`	Categorical distributions
`geom_boxplot()`	Outlier detection
`count()`	Frequency tables
`is.na()` / `sum(is.na())`	Missing data check

Before next class

📖 **Read:**

- R4DS Ch 10: Exploratory data analysis (sections 10.5–10.6)

✅ **Practice:**

- Explore at least 3 variables in `bfi`
- Look for outliers and missing data
- Start generating questions

Key takeaways

1. **EDA is an attitude** — curiosity, not confirmation
2. **Always visualize** — summary stats hide patterns
3. **Outliers are information** — investigate, don't delete
4. **Check missing data first** — it affects everything
5. **Ask questions, then answer them** — that's the loop

The one thing to remember

EDA isn't a step you finish. It's the habit of looking at your data before you believe anything about it.

Next time: EDA — Covariation

Get a head start

Assignment 4: Part 2 (Distributions)

Open Assignment 4 and start **Part 2**. You have everything you need:

1. **2.1** — Histogram of penguin body mass. Try ≥3 binwidths in your scratch work; submit the one that best shows the shape, with a comment explaining why.
2. **2.2** — Overlapping density plots of body mass, colored by species, with transparency.
3. **2.3** — Boxplot of body mass by species, then violin plot. Comment on what each shows that the other doesn't.

Due **Sunday, May 3 at 11:59 PM**.